

# ML Hackathon Evaluation Rules & Guidelines

Official Rules, Scoring Rubrics & Model Submission Protocol

## 1. Competition Scoring Rubric (100 Points Total)

| Category                             | Points | Description & Evaluation Metric                                                               |
|--------------------------------------|--------|-----------------------------------------------------------------------------------------------|
| Final Detection Performance          | 40     | Evaluated on hidden final_test.csv using F1-Score & PR-AUC. Plain accuracy is not used.       |
| Unknown / Shifted Attack Performance | 20     | Detection efficacy on unseen attack families and shifted network environments.                |
| Cost-Sensitive Loss Optimization     | 15     | Performance under asymmetric operational loss function (penalty weights announced at Hour 3). |
| Technical Approach & Analysis        | 10     | Feature engineering rigor, validation methodology, and verified model reproducibility.        |
| Robustness & Failure Analysis        | 5      | Understanding of model vulnerabilities, edge cases, and boundary conditions.                  |
| Final Presentation                   | 10     | 3-5 min technical defense answering the 8 required evaluation questions.                      |
| Total Score                          | 100    | Comprehensive Hackathon Score                                                                 |

## 2. 5-Hour Competition Timeline Progression

| Hour   | Phase           | Objectives & Deliverables                                                                         |
|--------|-----------------|---------------------------------------------------------------------------------------------------|
| Hour 1 | Find the Signal | Explore train.csv and validation.csv. Establish working baseline model and metrics.               |
| Hour 2 | Build the IDS   | Feature engineering (ratios, logarithms), preprocessing, algorithm tuning, ensembling.            |
| Hour 3 | Cost of Failure | Dynamic requirement change: Security team introduces operational risk inject and asymmetric loss. |
| Hour 4 | Unknown Attack  | Challenge dataset released (10% unlabelled). Adapt to shifted traffic and unfamiliar threats.     |
| Hour 5 | Final Incident  | Final test dataset released (15% unlabelled). Final submissions and 3-5 min technical defense.    |

## 3. Submission Format & Verification Rules

Submissions must be valid CSV files with **row\_id,prediction,probability**.

- row\_id**: 1-based sequential line number (1 to N) matching the test file exactly.
- prediction**: Binary integer (0 = Normal, 1 = Attack). Required.
- probability**: Float in [0.0, 1.0]. Recommended for PR-AUC scoring.
- Exactly 254,005 rows for challenge.csv; exactly 381,007 rows for final\_test.csv.

## 4. Model Submission & Reproducibility Guidelines (Mandatory at Hour 5)

Teams must submit their trained model package so organizers can independently verify model reproducibility. Package structure: **team\_name\_model/** containing:

- model.joblib** (or model.pkl): Serialized model pipeline and encoders.
- predict.py**: Standard inference script runnable via: `python predict.py --input test.csv --output pred.csv` (see template in starter/predict.py).
- requirements.txt**: Package dependencies (e.g. scikit-learn, pandas, numpy).
- metadata.json**: Team name, model architecture, and chosen decision threshold.

*Note: Organizers will test predict.py against hidden data. Verified reproducibility (>= 99.9% match) is required for full Technical Approach points.*